GuardLM: A Single-GPU Alignment Pipeline for Structured Content-Safety Risk Detection in Chinese

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Abstract

Industrial content moderation still leans on keyword blocklists that are trivially evaded by homophone, pinyin, and character-splitting variants, while general-purpose LLMs do not reliably emit the structured verdicts that downstream moderation systems require. We present **GuardLM**, a 1.7B-parameter Chinese content-safety classifier built end-to-end on a single RTX 4090 (24 GB). GuardLM couples (i) an adversarial data engine that unifies three public corpora at the 122k-text scale, fills scarce risk categories through scenario-matrix-constrained LLM synthesis, and manufactures 15.6k evasion variants; (ii) a three-stage alignment stack — QLoRA SFT, DPO on preference pairs derived from error attribution, and GRPO with a four-component reward — that raises overall F1 from 0.862 to 0.874 on a three-tier, leakage-checked benchmark; and (iii) a vLLM serving layer with JSON-schema-constrained decoding that pins the format error rate at 0.00% with no measured throughput tax. Our analysis shows that DPO mostly *migrates* errors between misses and false alarms rather than reducing them, whereas GRPO with an elevated sampling temperature (effective gradient groups 33%→50%) breaks this trade-off and improves both ends simultaneously (adversarial recall +1.6 pt, false-positive rate 3.57%→2.86%). A deployment study quantifies serving trade-offs: AWQ 4-bit quantization costs 4.81-pt F1, concentrated in adversarial recall (homophone −18.2 pt), so we retain bf16 serving, measured at 62.3 req/s under concurrency 16 with P99 latency 266 ms.

1 Introduction

Chinese online platforms moderate comments, danmaku, and private messages against several risk categories: pornography, insults, advertising diversion, and illegal content. Two deployment constraints shape the problem. First, *lexical* defenses are brittle: a word blocklist can be evaded at near-zero cost by homophone substitution (*xiéyn*), pinyin initials, or character splitting (*chizì*), which real users employ routinely. Second, *general-purpose* LLMs, although semantically capable, do not honor a fixed output contract: in our measurement, the untuned Qwen3-1.7B base model produced schema-compliant JSON for 0% of inputs, emitting free-form risk commentary instead of the machine-parseable verdicts a rule engine consumes.

GuardLM closes this gap with a 1.7B specialized model that is deliberately small enough to train and serve on one consumer GPU. The full stack — data engine, three-stage alignment, leakage-checked evaluation, and serving — is designed, measured, and documented as a reproducible vertical-domain pipeline.

Our contributions are:

- **An adversarial data engine** that unifies COLD, Safety-Prompts, and CValues into a 122k-text pool, fills acutely scarce categories via a 20-cell scenario matrix that constrains LLM synthesis (porn 38→156, ads 27→201), and generates 15.6k evasion variants with five deterministic augmentation families, with MinHash deduplication and provenance tracking guaranteeing zero train–eval leakage.
- **A three-stage alignment recipe** (QLoRA SFT → DPO → GRPO) on Qwen3-1.7B, with an *evaluation-first* harness, error attribution that converts benchmark errors into preference pairs, and a temperature probe that revives GRPO group diversity after DPO sharpening.
- **An error-migration analysis** showing DPO's net effect is internal error migration (miss ↔ false alarm) with a flat total error count, while GRPO simultaneously improves

adversarial recall and false-alarm rate — breaking the trade-off that DPO could not.

- **A deployment study** covering JSON-schema constrained decoding (format errors 0.2%→0.00%, zero throughput tax), single-GPU capacity baselines (62.3 req/s @ concurrency 16), and a quantization trade-off analysis leading to a data-driven rejection of AWQ 4-bit for the primary serving path.

2 Related Work

Chinese safety corpora and moderation. Public resources cover complementary slices of the safety problem: COLD (Yang et al., 2022) annotates offensive comments, Safety-Prompts (THU-CoAI, 2023) collects adversarial safety scenarios, and CValues (X-PLUG, Alibaba Group, 2024) contributes safety-contrast question–answer pairs. None individually matches a production taxonomy of platform risks, and none covers graphonym-based evasion at scale; our data engine unifies them and manufactures the missing adversarial surface.

Alignment methods. RLHF with PPO (Schulman et al., 2017) aligns models via a learned reward but is engineering-heavy. DPO (Rafailov et al., 2023) reframes preference optimization as a closed-form classification loss, and GRPO (Shao et al., 2024) replaces the critic with group-relative advantages, enabling outcome-rewarded RL within the memory budget of a single GPU. We use both in sequence and analyze *where each moves errors* rather than only aggregate scores.

Efficient fine-tuning and serving. LoRA (Hu et al., 2022) and QLoRA (Dettmers et al., 2023) make full alignment stacks feasible on consumer GPUs; LLaMA-Factory (Zheng et al., 2024) and TRL (von Werra et al., 2020) provide the training scaffolds. For serving, vLLM (Kwon et al., 2023) introduced paged attention for high-throughput inference, XGrammar (Dong et al., 2024) provides schema-constrained decoding, and AWQ (Lin et al., 2024) offers 4-bit activation-aware quantization. We measure the accuracy cost of quantization specifically for a small vertical-domain classifier, where, as we show, the damage concentrates in exactly the adversarial capabilities the model exists to provide.

3 Task Formulation

Given an input text x (a comment, danmaku, or private message), the model must emit

$$y = \{\texttt{"label"} : \ell,\ \texttt{"risk"} : r,\ \texttt{"reason"} : s\}, \tag{1}$$

where $\ell \in \{\texttt{porn}, \texttt{insult}, \texttt{ads}, \texttt{illegal}, \texttt{none}\}$, $r \in [0, 1]$ is a calibrated confidence, and s is a one-line attribution. The contract is deliberately minimal: three fields are sufficient for a downstream rule engine to branch on, while the `reason` field keeps individual decisions auditable by human reviewers.

We treat the task as five-class classification with a per-tier false-positive discipline: in content safety, a *miss* (risk text released) is operationally costlier than a *false alarm* (benign text held), so we report recall on risk classes and the false-positive rate (FPR) on `none` alongside F1.

4 Data Engine

4.1 Corpus unification

We ingest three public sources: COLD (31,692 offensive-comment annotations), Safety-Prompts (89,997 adversarial scenarios across 10 risk families), and CValues (38,689 safe/unsafe answer pairs, used in M3 for preference construction). After schema unification to our five-label taxonomy, normalization, and filtering, the pool holds 122,183 texts; MinHash deduplication (Broder, 1997) removes 33 fuzzy duplicates, leaving 122,150.

4.2 Scenario-matrix synthesis

The unified pool is severely imbalanced toward `insult` and `illegal`: before synthesis it contains only 38 `porn` and 27 `ads` training texts. Rather than scraping — legally and operationally fraught — we drive a commercial instruction-tuned LLM through a **20-cell scenario matrix**: {category} × {surface: comment / danmaku / private message / broadcast signal} × {style: colloquial / formal / coded}, with temperature-diverse prompting and a JSON-only response contract, in the spirit of Self-Instruct (Wang et al., 2023). Each candidate is human-reviewed; 494 of 495 survived, raising `porn` to 156 (4.1×) and `ads` to 201 (7.4×).

4.3 Adversarial augmentation

A five-family deterministic rewriter manufactures the evasion surface that public datasets lack: ho-

Component	Count	Note
COLD texts	31,692	offensiveness
Safety-Prompts	89,997	10 risk families
CValues pairs	38,689	preference source
Unified pool	122,183	5-label schema
After MinHash dedup	122,150	−33 near-dupes
Synthesized	494	reviewed; porn 4.1×
Adversarial variants	15,627	5 families
SFT instructions	12,148	risk:none 1:3
DPO preference pairs	2,176	error attribution
GRPO prompt pool	2,400	attack-weighted
Benchmark T1/T2/T3	250/250/150	leakage-checked

Table 1: Data engine statistics.

mophone substitution, pinyin abbreviation, character splitting, emoji interleaving, and traditional-Chinese conversion. Each variant carries an `origin_id` linking back to its seed. Augmentation yields 15,627 variants; training data mixes original and augmented texts at a 1:1 per-source ratio so that the model sees each attack pattern thousands of times while never losing the natural-text anchor.

4.4 Leakage control and benchmark

A naive pipeline risks evaluating on augmented neighbors of training texts. We enforce three controls: (i) MinHash near-duplicate removal across the full pool before split; (ii) split-first-then-augment ordering, so a variant never crosses the train/eval boundary; (iii) an automated leak check asserting zero intersection. The held-out benchmark is three-tier, 650 samples total: **T1** (250 conventional), **T2** (250 adversarial: six perturbation families, n per family ranging 10–120), and **T3** (150 boundary cases: slang, coded, and euphemistic risks near the none boundary).

5 Alignment Pipeline

5.1 Evaluation-first harness

Before any training, we build a harness that runs batched inference, parses verdicts with a five-level JSON tolerance chain (strict parse → substring extraction → field-repair → heuristic label mapping → none-default), and emits per-tier Markdown reports. Running the untuned base model through it establishes the necessity baseline: 100% format violations and unusable classification, motivating instruction tuning as stage one.

5.2 Stage 1: SFT

We fine-tune Qwen3-1.7B (Qwen Team, 2025) with QLoRA (Dettmers et al., 2023) on 12,148 instructions at a deliberate risk:none ratio of 1:3 — the untuned base model exhibits a strong risk prior (86–96% of texts judged risky), and negative reweighting is required to anchor the decision boundary. Training: rank 16, α 32, lr 10^{-4}, effective batch 32, 3 epochs — 1,083 steps in 12 min 16 s on one RTX 4090 (47.0 samples/s, 17.3 GB peak VRAM), eval loss 0.0056 with no divergence. Result: format compliance 0%→99.8%, overall F1 0.862, adversarial-tier (T2) recall 0.800.

5.3 Stage 2: DPO from error attribution

We bucket the 134 SFT errors into four causes: boundary misjudgment 42.5%, adversarial failure 38.8%, category confusion 17.9%, format 0.7%. Each error becomes a preference pair (chosen: the correct verdict with a faithful reason; rejected: the model's actual error), yielding 2,176 pairs. A first run at lr 5×10^{-6} never left the $\ln 2$ loss floor; lr 10^{-5} learns but pushes FPR to a 3.57% red line; adding 176 mirrored `none` pairs (chosen: none, rejected: over-blocked verdict) recovers the false-alarm direction while keeping boundary gains, adopted as dpo-v2 ($\beta = 0.1$).

5.4 Stage 3: GRPO with a temperature probe

A GRPO (Shao et al., 2024) dry-run exposed a pathology inherited from DPO: the sharpened policy produced zero-advantage groups for 75–100% of prompts (entropy 0.006), which starves the group-relative objective of gradient signal. A temperature probe over $\{1.0, 1.2, 1.5\}$ showed $T = 1.5$ raises effective (non-degenerate) groups from 33% to 50% with zero format degradation — diagnosis before tuning, not blind hyperparameter search. Final run: $T = 1.5$, group size 8, $\beta_{\mathrm{KL}} = 0.04$, lr 5×10^{-6}, one epoch over a 2,400 adversarial-weighted prompt pool, 300 steps in 24.8 min.

5.5 Reward design

The reward has four components, each guarding one failure mode: *correctness* (2.0, with incremental partial credit for near-miss categories) dominates; *format* (1.0) locks the JSON contract; *adversarial* boost ($\times 1.5$) concentrates learning on evasion texts; and a *length penalty* suppresses the classic RL output-inflation failure. Section **??** shows all four behaved as designed.

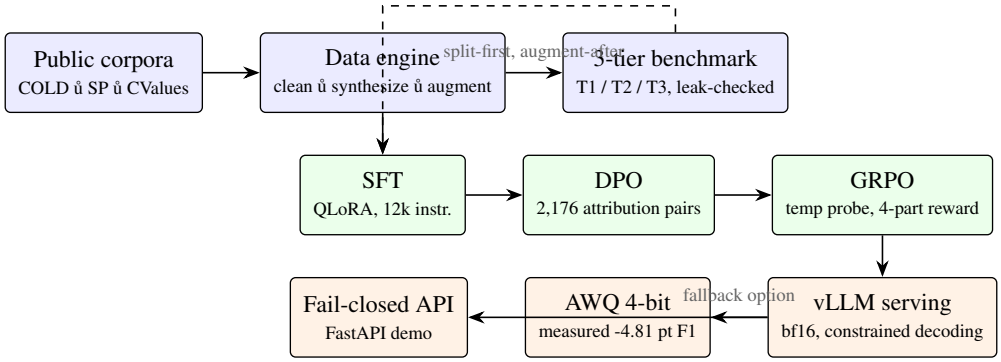


Figure 1: GuardLM pipeline. The benchmark feeds every stage through error attribution; the serving layer measures three deployment forms under one protocol.

6 Experiments

6.1 Setup

All evaluation uses the 650-sample three-tier benchmark under a fixed protocol: temperature 0.7, top-p 0.8, top-k 20, max 64 new tokens, no chain-of-think mode; parse failures count as none (a held verdict), and the format error rate is reported separately. Iteration decisions are additionally protected by McNemar and Fisher exact tests on per-sample flips, with a pre-registered 2-pt red line beyond the ± 2 pt noise band of a 650-sample benchmark.

6.2 Main results

Table 2 tracks the four training stages plus two serving forms. Overall F1 rises monotonically $0.000 \rightarrow 0.862 \rightarrow 0.867 \rightarrow 0.874$, and the vLLM bf16 deployment preserves it (0.8755, within the noise band, confirming zero serving loss). T3 (boundary) F1 improves $0.805 \rightarrow 0.828$ across SFT→DPO, then holds within noise at GRPO (0.819; McNemar $p = 0.688$ on the tier).

6.3 DPO migrates errors; GRPO breaks the trade-off

DPO's headline gain (T3 +2.3 pt) hides its net effect: comparing per-sample outcomes across the benchmark, the *total* error count is flat — errors migrate between misses and false alarms rather than disappear, and FPR crosses the red line (0.71%→3.57%). A KL check ($\mathrm{KL}(\pi \| \pi_{\mathrm{ref}}) \approx 0.001$ bits/token) rules out policy drift as the cause, pointing instead to the preference signal itself rebalancing along a single trade-off axis.

GRPO acts on the same benchmark with a different signature: 19 errors repaired vs. 10 regressed (net +9, McNemar $p = 0.136$ overall; $p = 0.092$ on T2, marginally significant) *while simultaneously* lowering FPR 3.57%→2.86% and raising adversarial recall 0.812→0.828 — both ends improve, which DPO could not achieve on this budget. Flip inspection shows the mechanism: repaired cases are adversarial `illegal/insult` misses, while the mirrored-none direction from the reward's partial credit restores over-blocked benign texts.

6.4 Training dynamics

Figure 2 shows the three training stages from event-log scalars. SFT converges smoothly; DPO's loss separates from $\ln 2$ only at the corrected learning rate, with margins growing to 0.214; GRPO's reward climbs while the zero-advantage fraction falls to 0.125 — direct evidence that the temperature probe fixed the group-diversity pathology. Figure 3 shows the run stayed healthy end-to-end: KL bounded, entropy away from zero, completions short and stable.

6.5 Ablation: continued pre-training

A controlled ablation adds domain continued pre-training (CPT, 109k documents, 70% risk text) before the identical SFT recipe (sft-v2). Domain slang generalization improves sharply: the P5 argot probe F1 0.824→0.947 (+12.4 pt), overall F1 0.898, adversarial recall 0.868. But the risk-heavy corpus re-introduces the risk prior on the pre-training side: FPR rises 0.71%→9.29% — the miss/false-alarm trade-off resurfaces at a deeper layer. Catastrophic-forgetting probes (chitchat, commonsense, arithmetic) all pass, and the experiment achieves its ablation purpose, but production keeps grpo-v1; replay-mixed CPT remains the open compromise.

Stage	Model	Overall F1	T1 F1	T2 F1	T2 Recall	FPR	Format err.
base	Qwen3-1.7B (untuned)	0.000*	–	–	–	–	100%
SFT	sft-v1	0.8619	0.8731	0.8889	0.8000	0.71%	0.15%
DPO	dpo-v2	0.8669	0.8500	0.8962	0.8120	3.57%	0.15%
GRPO	grpo-v1	**0.8740**	0.8713	**0.9059**	**0.8280**	**2.86%**	**0.00%**
serving	guardlm-final (vLLM bf16)	0.8755	0.8780	0.9011	0.8200	3.57%	0.00%
serving	guardlm-awq (vLLM 4-bit)	0.8274	0.8469	0.8558	0.7480	2.14%	0.00%

Table 2: Main results on the 650-sample three-tier benchmark. *The untuned base does not produce schema-compliant JSON (0% compliance), so F1 is undefined; the row documents the necessity of instruction tuning. Serving rows reuse grpo-v1 weights under identical protocol.

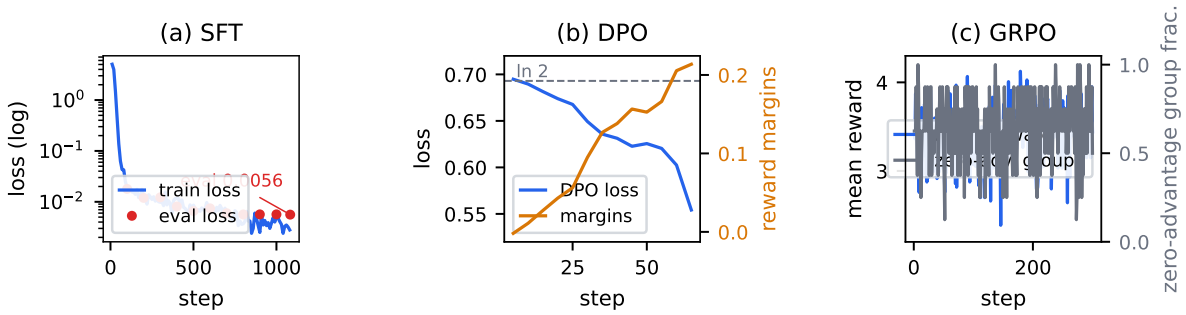


Figure 2: Alignment-stage training curves from TensorBoard event logs. (a) SFT: train/eval loss, 1,083 steps in 12 min; eval loss bottoms at 0.0056 with no overfitting fork. (b) DPO: loss escapes the $\ln 2$ floor as reward margins separate. (c) GRPO: mean reward rises 2.39→4.13 while zero-advantage group fraction collapses 1.0→0.125 after the temperature probe.

Deployment	F1	Recall	Weight
HF transformers (bf16)	0.8740	0.7824	3.3 GiB
vLLM (bf16)	0.8755	0.7863	3.3 GiB
vLLM (AWQ 4-bit)	0.8274	0.7098	1.3 GiB

Table 3: Three deployment forms of grpo-v1 weights, identical 650-sample protocol. AWQ trades 4.81-pt F1 (Recall −7.65 pt) for 61% smaller weights and 2.6× faster batch inference (183 s→69 s).

Perturbation	n	bf16 R	AWQ R
homophone	22	90.9	72.7
split	10	90.0	80.0
sp_attack	120	73.3	64.2
emoji	25	96.0	92.0
pinyin	26	76.9	73.1
traditional	47	93.6	93.6

Table 4: T2 adversarial recall by perturbation type, bf16 vs. AWQ (%). Character-level evasions collapse under 4-bit noise; semantic-level evasions survive.

6.6 Ablation: 4-bit quantization

Table 3 compares three deployment forms under one protocol. vLLM bf16 costs nothing (F1 +0.15 pt, within noise). AWQ 4-bit costs 4.81 pt F1, and Table 4 localizes the damage: character-level evasions (homophone −18.2 pt, split −10.0 pt, sp_attack −9.1 pt) collapse while semantic-level perturbations (traditional, emoji) are nearly unaffected. Confusion-matrix deltas confirm the mechanism: false negatives rise 109→148 while false positives fall — quantization flattens boundary logits and shifts the whole decision mass toward none. For a 1.7B model whose weight redundancy is already thin, AWQ's 1% salient-weight protection does not cover the discriminative region this task lives in. Because a miss is costlier than a false alarm in content safety, we reject AWQ for the primary path and keep it as a measured fallback for ≤8 GB-VRAM or offline pre-screening deployments.

7 Deployment

Constrained decoding. The serving layer turns format assurance from a post-hoc check into a decoding-time guarantee: vLLM's `json_schema` guided decoding (XGrammar backend) masks logits against the output contract throughout generation. Measured effect: format errors 0.2%→0.00% under the serving protocol, at zero throughput tax — structured decoding on/off leaves request rate unchanged and costs −0.17 pt F1, inside the noise

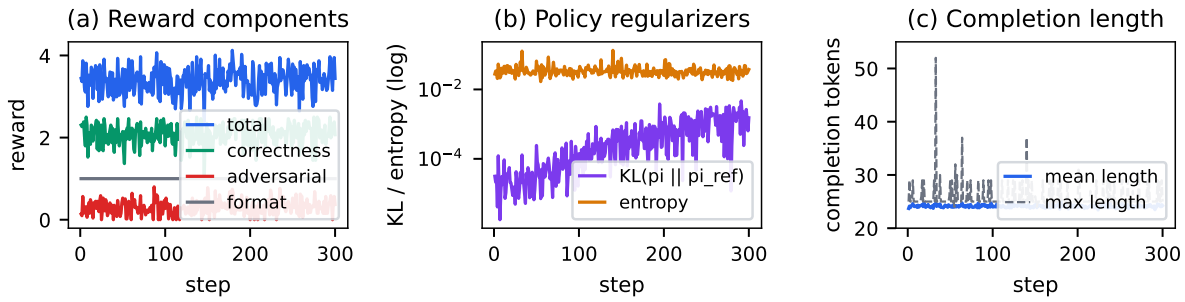


Figure 3: GRPO training health over 300 steps. (a) Reward components: correctness dominates and rises; the adversarial bonus activates on evasion prompts; format stays saturated. (b) KL to the reference stays below 5×10^{-3} and entropy remains in 0.017–0.134 (log scale) — mild policy movement, no collapse. (c) Completion length holds at 23–25 tokens (max 52): zero RL length drift, validating the length-penalty component.

Concurrency	req/s	P99 E2E	TTFT P99
1	28.2	35 ms	27 ms
4	49.5	81 ms	45 ms
16	62.3	266 ms	74 ms

Table 5: Single-GPU serving capacity (bf16, constrained decoding on, stable in-flight load).

band. The five-level parse fallback in the demo layer remains as defense in depth.

Capacity. Table 5 reports the steady-state capacity baseline on one RTX 4090 with prefix caching disabled (kept off so the numbers are reproducible): 62.3 req/s at concurrency 16 with P99 end-to-end latency 266 ms — comfortably within interactive moderation budgets.

Fail-closed discipline. The FastAPI gateway follows a fail-closed contract: if the inference backend is unreachable or a verdict cannot be parsed, it returns an explicit 502 rather than a fabricated verdict. In content safety, silently defaulting to none on backend failure would widen the miss rate exactly when the system is degraded; an explicit refusal routes the text to a fallback queue instead.

8 Conclusion

GuardLM demonstrates that a complete vertical-domain risk-identification stack — adversarial data engine, three-stage alignment, leakage-checked three-tier evaluation, and quantitatively characterized serving — fits on a single consumer GPU while reaching 0.874 overall F1 and 0.00% format errors in production form. Two findings travel beyond this domain: (i) DPO applied to a classifier trades misses against false alarms without shrinking total error, while GRPO at a probed sampling temperature improves both ends at once; and (ii) post-training quantization of a small specialized classifier silently destroys exactly the adversarial capabilities it was built for, so the bf16/quantized decision must be measured, not assumed. The data engine, harness, training recipes, and deployment trade-offs are documented for reproduction.

Limitations

The benchmark is 650 samples; improvements within ±2 pt are inside the noise band, and we used significance tests rather than raw deltas for all iteration decisions, but a larger tier-3 set would sharpen boundary conclusions. All results use one base model (Qwen3-1.7B) and Chinese only; cross-lingual transfer of the augmentation families is untested. Synthesis relied on a commercial LLM API whose exact model changed over the project's lifetime; the 494 surviving samples are human-reviewed, but the approach inherits any upstream alignment biases. CPT ablation suggests replay-mixed continued pre-training could resolve its FPR regression, but this remains future work, as do quantization-aware fine-tuning alternatives (GPTQ-INT8, QAT) that might close the 4.81-pt AWQ gap.

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